



Cape Peninsula
University of Technology

APPLICATIONS DEVELOPMENT THEORY 4

SUBJECT GUIDE FOR:	2025
COURSE CODE:	APT470S

Agile Methods Research, Analysis & Video Presentation

Instructions:

- i. Total Marks: **100**
- ii. Answer all questions.
- iii. This is an **individual assignment**, but collaboration and discussion with colleagues are encouraged to enhance understanding and insights.
- iv. Solutions must be uploaded to Blackboard by the Close of Business (COB) on **Friday, March 13, 2026**.

Assignment Overview

This research assignment aims to deepen your understanding of Agile methodologies through active investigation, critical analysis, and professional presentation. You will work individually but may consult your group colleagues to become a subject-matter expert using an Agile approach and share your findings in a structured video presentation.

Assignment Structure

Component	Description	Marks
Research & Analysis	Depth of investigation and critical evaluation	65
Presentation Slides	Quality, structure, citations, and visual effectiveness	20
Video Presentation	Individual presentation with video	10
Others	Such as evidence of class group discussion, uploaded articles, etc	05
Total		100

Research Questions

You have been assigned an Agile methodology (Extreme Programming) for your research. Your assignment must address ALL of the following questions:

1. Research & Analysis (65 Marks)

Extreme Programming (XP)

No.	Research Question	Marks
1	Trace the origins of XP through the work of Kent Beck, Ward Cunningham, and Ron Jeffries. What was the "Chrysler Comprehensive Compensation" (C3) project's role in XP's development?	10
2	Analyze the five values of XP: Communication, Simplicity, Feedback, Courage, and Respect. How do these values manifest in the technical practices of XP?	15
3	Critically evaluate the core XP practices: Test-Driven Development (TDD), Pair Programming, Collective Ownership, Continuous Integration, On-site Customer, and Refactoring. Which practices have the strongest empirical support?	10
4	Examine the controversial practices of XP: Pair Programming and On-site Customer. What research exists on the productivity impact of pairing? How has remote work challenged the "on-site" requirement?	10
5	Compare XP's technical rigor with other Agile methods that place greater emphasis on management and planning. Why is XP considered the "engineering" Agile method?	10
6	Investigate the evolution of XP since its formalization in the late 1990s. Which practices have endured, which have been adapted, and why?	10

Deliverable Requirements

2. Research Slides (20 Marks)

Create a cohesive individual presentation (10-12 slides, including opening and concluding slides) that addresses ALL research questions for your assigned method. The slides must:

- **Structure:** Include a title slide, your student ID, and contents organised in a logical order.

- **Content:** Demonstrate critical analysis, not mere description. Engage with academic literature (minimum of 8 good peer-reviewed sources).
- **Design:** Follow professional presentation standards, consistent formatting, appropriate visuals, clear typography, and effective use of diagrams
- **Citations:** Use APA 7th edition for all in-text citations and reference list

3. Video Presentation (10 Marks)

Create a single compiled video (10 minutes) where:

- **Presentation:** You MUST individually present your work for ALL the research questions.
- **Video-in-Video Format:** You MUST record yourself (head and shoulders, facing the camera) and insert this video recording into the bottom right corner of all your slides

Marking Rubric

Research & Analysis (35 Marks)

Criteria	Excellent (85-100%)	Good (70-84%)	Satisfactory (60-69%)	Needs Improvement (<60%)
Depth of Research (20)	Engages with 10+ high-quality academic sources; demonstrates a comprehensive literature review	Engages with 8 academic sources: good coverage of literature	Engages with 5-7 sources: adequate but limited depth	Fewer than 5 sources; superficial engagement
Critical Analysis (25)	Evaluates competing perspectives; identifies strengths/weaknesses; draws original insights	Good analysis with some evaluation; mostly balanced	Descriptive rather than analytical; limited critique	Purely descriptive; no critical evaluation
Method-Specific Understanding (20)	Demonstrates sophisticated grasp of concepts, tensions, and practical implications	Strong understanding of core concepts with some ideas	Accurate but surface-level understanding	Factual errors or misconceptions

Presentation Slides (20 Marks)

Criteria	Excellent (85-100%)	Good (70-84%)	Satisfactory (60-69%)	Needs Improvement (<60%)
Structure & Flow (5)	Logical progression; clear signposting; seamless transitions	Good structure with minor flow issues	Adequate structure, but some disjointed sections	Poor organization; difficult to follow
Visual Design (5)	Professional, consistent, visually engaging; effective diagrams and graphics	Good design with minor inconsistencies	Basic design; text-heavy in places	Poor design; distracting or unprofessional
Clarity & Conciseness (5)	Information presented with exceptional clarity; appropriate depth	Clear with some verbosity or oversimplification	Understandable but unclear in places	Unclear, confusing, or overly dense
Referencing (5)	Perfect APA 7th formatting; comprehensive reference list	Good referencing with minor formatting errors	Adequate referencing with several errors	Poor or missing citations
